Self-similar cubic checkpoint

The self-similar polarized theorem for

$P_C(x,y) = (x-y)^3 + C(x+y)$

Status

The self-similar assertion **SS-BP is true**. In fact it holds with the same explicit constant supplied by the frozen homogeneous two-prime argument.

Throughout, $q \neq r$ are fixed odd primes,

$$\lambda = qr, \qquad M_t = q^{3t}r^{3t} = \lambda^{3t}, \qquad C_t = q^{2t}r^{2t} = \lambda^{2t}.$$

For $p \in \{q, r\}$, put

$$A_p = (1 - p^{-1})^{-1}$$

and define the constant from the frozen triple-convolution lemma by

$$\mathcal{C}_{q,r} = 6(A_q A_r^2 + A_q^2 A_r). \tag{1}$$

The main result of this checkpoint is the following.

Theorem (SS-BP). For every $t \geq 1$ and all nonempty $A, B \subseteq \mathbb{Z}/M_t\mathbb{Z}$,

$$\boxed{|P_{C_t}(A,B)| \geq b_{q,r} \frac{|A||B| \max\{|A|,|B|\}}{M_t^2}, \qquad b_{q,r} = \frac{1}{36\mathcal{C}_{q,r}}.} \tag{2}$$

No internet or external source was used.

A. Frozen checkpoint confirmation

The six frozen intermediate-singular conclusions in the research prompt were used as assumptions and were not re-proved. One precise internal lemma from the frozen homogeneous two-prime proof is also used below: the audited triple-convolution/maximal-cross-edge inequality stated in (19). That is the quantitative engine which originally proved the frozen homogeneous polarized theorem.

The new work is the weighted anchored-root estimate in Lemma 1. It controls all center translations at once and turns the frozen homogeneous charging inequality into (2).

The only general standard facts used are:

1. the Chinese remainder theorem;
2. the elementary fact that a degree-three polynomial over a field has at most three roots;
3. elementary p-adic valuation arithmetic and the factorization $X^3 - Y^3 = (X-Y)(X^2+XY+Y^2)$.

For the optional collision example in Section C, we also use the following standard finite-unit-group fact: for an odd prime p, the group $(\mathbb{Z}/p^n\mathbb{Z})^\times$ is cyclic. Consequently cubing is an automorphism on $1+p\mathbb{Z}/p^n\mathbb{Z}$ when $p \neq 3$, while for $p = 3$ its image on $1 + 3\mathbb{Z}/3^n\mathbb{Z}$ is $1 + 9\mathbb{Z}/3^n\mathbb{Z}$.

B. Verdict on the self-similar theorem

The verdict is positive: (SS-BP) holds uniformly in t, A, B, with the explicit constant in (2).

The scale is sharp. For every residue a, take

$$A = B = a + \lambda^t(\mathbb{Z}/M_t\mathbb{Z}).$$

Then $|A| = |B| = \lambda^{2t}$, while both the variable cubic term and the variable part of the linear term vanish modulo M_t. Hence

$$P_{C_t}(A, B) = \{2C_t a\}, \qquad \frac{|A||B|\max(|A|, |B|)}{M_t^2} = 1.$$

Thus the polarized cubic exponent cannot be improved even in this exact self-similar family.

The proof does **not** rely on a false assertion that hierarchical translation addresses make all transverse blocks disjoint. Incomparable blocks can have unbounded, and even exact, overlap. Instead, the proof keeps one input as an anchor. Once y is fixed,

$$P_{C_t}(y + u, y) = u^3 + C_t u + 2C_t y, \tag{3}$$

and once x is fixed,

$$P_{C_t}(x, x - u) = u^3 - C_t u + 2C_t x. \tag{4}$$

The translations $2C_t y$ and $2C_t x$ only change the right-hand side of a one-variable congruence. At the exact 2:3 scaling, every weighted fiber of the anchored cubics $u^3 \pm C_t u$ has uniformly bounded mass. This is the missing translated charging principle.

C. Translation-address structure

C.1 Exact recursion

Suppose $0 \le d \le t$ and x, y have a common base-λ prefix $w \pmod{\lambda^d}$:

$$x = w + \lambda^d X, \qquad y = w + \lambda^d Y.$$

A direct expansion gives

$$\boxed{P_{C_t}(x, y) = 2C_t w + \lambda^{3d} P_{C_{t-d}}(X, Y).} \tag{5}$$

If the common digits are $a_0, \dots, a_{d-1}$, so that

$$w = a_0 + \lambda a_1 + \cdots + \lambda^{d-1} a_{d-1},$$

then the accumulated translation is

$$2C_t w = C_t(2a_0 + \lambda(2a_1) + \cdots + \lambda^{d-1}(2a_{d-1})). \tag{6}$$

Thus the proposed hierarchical address is real and exact.

C.2 What addresses do separate

Along one ancestor-descendant chain, first-divergence side bands are disjoint. Indeed, at a depth-d divergence the residual difference is a unit at at least one of q, r. At that prime the residual polynomial in (5) is a unit, whereas every pair continuing into the common child has residual value divisible by that prime. After multiplication by λ^{3d}, the two output bands remain distinct.

This explains why an ordinary single-spine leakage construction does not by itself defeat (SS-BP).

C.3 What addresses do not separate

At a fixed depth d, the coarse containers

$$2C_t w + \lambda^{3d}(\mathbb{Z}/M_t\mathbb{Z})$$

for prefixes w, w' (mod λ^d) can intersect only if

$$\lambda^{3d} \mid 2C_t(w - w'),$$

or equivalently

$$\boxed{w \equiv w' \pmod{\lambda^{\max(0,3d-2t)}}.} \tag{7}$$

Hence the address-only multiplicity allowed by (7) is

$$\lambda^{d-\max(0,3d-2t)} = \begin{cases} \lambda^d, & d \le 2t/3, \\ \lambda^{2(t-d)}, & d > 2t/3. \end{cases} \tag{8}$$

This is unbounded near the critical depth $2t/3$.

The overlap is not merely an artifact of coarse containers. Let $t = 3H + 1$, $d = 2H$, and let w run through all residues modulo λ^d. By the finite-unit-group fact stated in Section A, there is a unit e_w modulo $\lambda^{3(t-d)}$ such that

$$e_w^3 \equiv 1 - 2\lambda^2 w \pmod{\lambda^{3(t-d)}}. \tag{9}$$

Indeed, the right side is congruent to 1 modulo p^2 at each $p \in \{q, r\}$. For $p \neq 3$, cubing is an automorphism on $1 + p^2\mathbb{Z}/p^n\mathbb{Z}$; for $p = 3$, its image on $1 + 3\mathbb{Z}/3^n\mathbb{Z}$ is $1 + 9\mathbb{Z}/3^n\mathbb{Z}$. The Chinese remainder theorem then supplies e_w.

Set, modulo M_t,

$$x_w = w + \frac{\lambda^d e_w}{2}, \qquad y_w = w - \frac{\lambda^d e_w}{2}.$$

The pair (x_w, y_w) first diverges at depth d, but

$$\begin{aligned} P_{C_t}(x_w, y_w) &= \lambda^{3d} e_w^3 + 2\lambda^{2t} w \\ &\equiv \lambda^{3d} \pmod{M_t}. \end{aligned} \tag{10}$$

Thus λ^{2H} different translation addresses yield the same actual output. The proposed unweighted bounded-overlap address lemma is false. The positive theorem must count all Cartesian cross-pairs rather than selected matched blocks.

D. Main new lemma: anchored weighted fibers

For a prime p, write

$$v_p(0) = 3t, \qquad \omega_{p,t}(u) = p^{-2\min(v_p(u),t)} \pmod{p^{3t}}. \tag{11}$$

Lemma 1 (self-similar anchored-root bound). Let p be an odd prime, let c be a unit modulo p, and let $\varepsilon \in \{1, -1\}$. For every $z \pmod{p^{3t}}$,

$$\boxed{\sum_{\substack{u \bmod p^{3t} \\ u^3 + \varepsilon p^{2t} cu = z}} \omega_{p,t}(u) \le 3.} \tag{12}$$

The estimate is uniform over the unit c; in the two CRT coordinates below, c is the unit contributed by the other prime and may depend on t.

Proof. First suppose $z = 0$. If $h = v_p(u) < t$, then

$$v_p(u^3 + \varepsilon p^{2t} cu) = 3h < 3t,$$

because $3h < 2t + h$. Hence such a u is not a root. Conversely every $u \in p^t\mathbb{Z}/p^{3t}\mathbb{Z}$ is a root. There are p^{2t} such elements and each has weight p^{-2t}, so their total weight is 1.

Now suppose $z \neq 0$, and put $e = v_p(z) < 3t$. Every root has a unique valuation $h < t$, and the same valuation comparison forces

$$e = 3h. \tag{13}$$

Write $s = t - h$ and $u = p^h v$, where v is a unit. After division by p^{3h}, the root congruence becomes

$$v^3 + \varepsilon p^{2s} cv \equiv z/p^{3h} \pmod{p^{3s}}. \tag{14}$$

Each solution $v \pmod{p^{3s}}$ has exactly p^{2h} lifts back to $u \pmod{p^{3t}}$.

There are at most three unit solutions of (14). If $p \neq 3$, reduction modulo p has at most three roots. Two roots with the same reduction cannot be distinct modulo p^{3s}, because their difference times

$$v_1^2 + v_1 v_2 + v_2^2 + \varepsilon p^{2s} c$$

is zero, and the displayed second factor is a unit modulo p.

If $p = 3$, all unit roots have the same residue modulo 3 because $v^3 \equiv v \pmod 3$. If $v_1 \equiv v_2 \pmod 3$, then

$$v_3(v_1^2 + v_1v_2 + v_2^2 + \varepsilon 3^{2s}c) = 1. \tag{15}$$

Indeed, writing $v_1 = v_2 + 3a$, the quadratic sum is $3(v_2^2 + 3av_2 + 3a^2)$, and the term $3^{2s}c$ is divisible by 9. The difference factorization therefore gives $v_1 \equiv v_2 \pmod{3^{3s-1}}$. A fixed class modulo 3^{3s-1} has only three lifts modulo 3^{3s}.

Thus the number of roots u of valuation h is at most $3p^{2h}$. Multiplication by their common weight p^{-2h} proves (12). □

The cancellation in the last line is the exact fiber-level renormalization: factoring out a root valuation h sends t to $t-h$, produces p^{2h} discarded lifts, and the weight contributes precisely p^{-2h}.

Under CRT, define

$$\Omega_t(u) = q^{-2\min(v_q(u),t)}r^{-2\min(v_r(u),t)}. \tag{16}$$

Lemma 1 factors between q and r, and therefore gives, for both signs,

$$\sum_{u^3 \pm C_t u = z} \Omega_t(u) \le 9. \tag{17}$$

Fixing one input in (3) or (4) now yields the genuinely translated statement.

Corollary 2 (anchored translated charging). For every output z,

$$\boxed{\sum_{\substack{(x,y)\in A\times B\\ P_{C_t}(x,y)=z}} \Omega_t(x-y) \le 9\min\{|A|,|B|\}.} \tag{18}$$

The important feature is the factor $\min(|A|,|B|)$. It is the retained anchor mass which the failed full-ring coarea estimate discarded.

E. Complete quantitative proof

Let $a = |A|$ and $b = |B|$. The exact audited product-tree estimate inside the frozen homogeneous proof is

$$\boxed{\sum_{x\in A, y\in B} \left(\frac{\operatorname{ord}(x-y)}{M_t}\right)^2 \ge \frac{a^2b^2}{4\mathcal{C}_{q,r}M_t^2}.} \tag{19}$$

Here $\operatorname{ord}(u)$ is the additive order of u in $\mathbb{Z}/M_t\mathbb{Z}$, with $\operatorname{ord}(0) = 1$. Under CRT,

$$\left(\frac{\operatorname{ord}(u)}{M_t}\right)^2 = q^{-2\min(v_q(u),3t)}r^{-2\min(v_r(u),3t)}. \tag{20}$$

Since truncation at t gives a larger weight than truncation at $3t$,

$$\Omega_t(u) \ge \left(\frac{\operatorname{ord}(u)}{M_t}\right)^2. \tag{21}$$

Consequently

$$\sum_{x\in A,y\in B}\Omega_t(x-y)\geq\frac{a^2b^2}{4\mathcal{C}_{q,r}M_t^2}. \tag{22}$$

On the other hand, group the same sum by the actual outputs of P_{C_t}. Corollary 2 gives

$$\sum_{x\in A,y\in B}\Omega_t(x-y)\leq 9\min(a,b)|P_{C_t}(A,B)|. \tag{23}$$

Combining (22) and (23), and using $a^2b^2/\min(a,b)=ab\max(a,b)$, gives

$$|P_{C_t}(A,B)|\geq\frac{1}{36\mathcal{C}_{q,r}}\frac{ab\max(a,b)}{M_t^2}.$$

This is (2). The constant depends only on the two fixed primes and is independent of t,A,B. No density range, integrality exception, or hidden depth-dependent loss remains.

F. Why the earlier obstructions are overcome

F.1 Static coarea

The forbidden strategy sought a set-independent weight whose mass on every **full two-variable output fiber** was bounded by an absolute constant. The weight Ω_t does not have that property, and no such claim is made. Instead, (18) bounds its mass on the restricted fiber $(A\times B)\cap P^{-1}(z)$ by $9\min(|A|,|B|)$. This bound comes from fixing an actual input and using the correlation $v=u+2y$ or $v=2x-u$. Replacing the true correlated (u,v)-set by a Cartesian product would destroy exactly this fact.

F.2 Mass-only recursion

The failed Bellman induction could only charge a side rectangle by its local cubic mass and therefore lost the third large factor. Inequality (19), coming from the frozen maximal-cross-edge/triple-convolution argument, already charges all Cartesian cross-pairs and supplies a^2b^2. The anchored fiber capacity in (18) removes only $\min(a,b)$, leaving the desired polarized quantity $ab\max(a,b)$. No depthwise multiplication of lossy child constants occurs.

F.3 Address-only charging

Addresses do separate ancestor-descendant side bands, but (10) shows that incomparable bands can overlap with multiplicity tending to infinity. The proof of (2) does not need to choose a prefix-free family or bound that raw overlap. The one-variable root bound controls the total weighted multiplicity of all translated branches simultaneously.

G. Consequences for full BP2+ and the three-prime theorem

The canonical self-similar branch is now closed positively. The same proof also applies, without a new idea, to the rectangular exact self-similar family

$$M = q^{3t_q} r^{3t_r}, \qquad C = c_0 q^{2t_q} r^{2t_r}, \qquad (c_0, qr) = 1, \tag{24}$$

with the same constant (2). Each local factor separately has the exact 2:3 relation used in Lemma 1.

This does **not** prove full BP2+ for arbitrary intermediate coefficients. The exact ratio is used at the critical shell. To see the obstruction to an automatic extension, let $p \geq 5$, use modulus p^{3h+N}, and take

$$C = -3p^{2h}, \qquad z = -2p^{3h}.$$

Writing $u = p^h w$ gives

$$u^3 + Cu - z = p^{3h}(w-1)^2(w+2).$$

For $w \equiv 1 \pmod{p^{\lceil N/2 \rceil}}$, this vanishes modulo p^{3h+N}. Modulo the residual p^N there are $p^{\lfloor N/2 \rfloor}$ such roots, and each has p^{2h} lifts in the original variable. Their total p^{-2h}-weight is at least

$$p^{\lfloor N/2 \rfloor}, \tag{25}$$

which is unbounded. In the self-similar core $N = 0$; this residual critical fiber is exactly what disappears.

Equation (25) is an obstruction to the present proof, not a counterexample to BP2+. Therefore the full arbitrary-C three-prime theorem remains open, because the frozen implication BP2+ $\Rightarrow$ that theorem still needs BP2+ outside the exact self-similar ratios. The many-prime/global real-set construction is likewise not closed by this checkpoint.

H. Exactly one next unresolved statement

For fixed distinct odd primes $q, r \geq 5$, determine whether there is a constant $b_{q,r} > 0$ such that, uniformly in $t \geq 1$ and all nonempty $A, B \subseteq \mathbb{Z}/M_t\mathbb{Z}$,

$$\boxed{\begin{array}{c} M_t = q^{4t} r^{3t}, \qquad C_t = -3q^{2t} r^{2t}, \\ |P_{C_t}(A,B)| \geq b_{q,r} \dfrac{|A||B| \max\{|A|,|B|\}}{M_t^2}. \end{array}} \tag{26}$$

This is the first canonical rectangular family in which one prime retains the solved 2:3 scaling while the other has a growing residual double-critical fiber of the form displayed in (25).